UG10096

FR1_FR2_Test_Tool DPD and Open-loop QEC User Guide

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User guide
CONFIDENTIAL

Document information

Information	Content
Keywords	UG10096, FR1, FR2, FR1_FR2_Test_Tool
Abstract	The document describes how to perform open-loop QEC calibration and RX IQ correction. It covers the detailed steps for closed-loop DPD training from time-domain and frequency-domain waveform sources.

1 Introduction

FR1_FR2_Test_Tool is a software tool running on the LA12xx device VSPA cores. It can:

- Generate 5G signals or single tone on LS-DCS for sub 6 GHz FR1 to test RF card performance
- Generate 5G signals or single tone on HS-DCS for mmwave FR2 to test RF card performance

The document describes how to perform open-loop QEC calibration and RX IQ correction. It covers the detailed steps for closed-loop DPD training from time-domain and frequency-domain waveform sources.

2 Environment setup

The following is a list of hardware and software components that are used for the setup.

- Baseband platform: LA1224-RDB-B-LS
- RF platform: Diora-3QB2 (N78) Diora-3QC (N79)
- BSP: LA1224-RDB BSP 2.4 / 3.0
- RF driver: Diora driver 2.1.1
  https://metanoia-comm.atlassian.net/servicedesk/customer/portal/3
- FR1_FR2_TOOL test tool script version: 4.0
- FR1_FR2_TOOL VSPA image version: 4.0.5
- Python: 3.6

Figure 1 depicts the block diagram of the setup used.

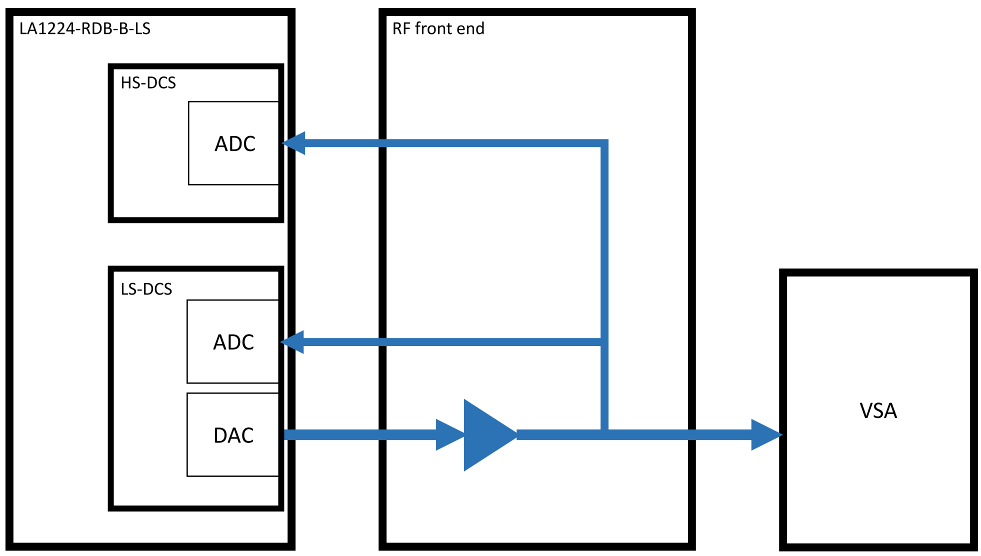

Figure 1. Block diagram

Figure 2 shows the LA1224-RDB-B-LS board and Diora RF cards used on the board.

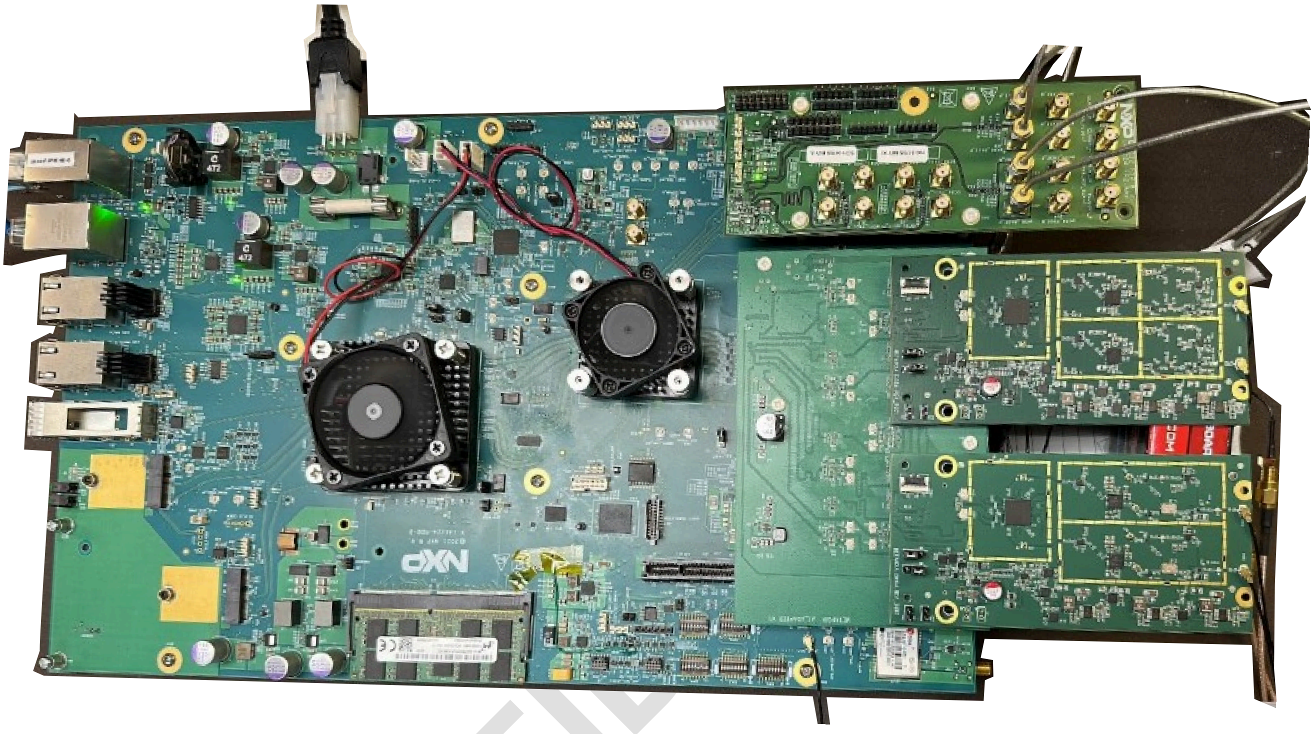

Figure 2. LA1224-RDB-B-LS and Diora RF cards

3 VSPA image prerequisites

The following list describes the prerequisites for the VSPA image.

- QEC calibration:
  No prerequisites required
- RX IQ correction:
  No prerequisites required
- DPD training: Time-domain waveform source
  ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
  – LS training: Observation on LS ADC to capture waveform at 245.76 Msps
  – HS training: Observation on HS ADC to capture waveform at 983.04 Msps
- DPD training: Frequency-domain waveform source
  ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD
  – LS training: Observation on LS ADC to capture waveform at 245.76 Msps
  ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
  – HS training: Observation on HS ADC to capture waveform at 983.04 Msps

Note: *The performance of DPD is affected by the signal quality, LO leakage, training sampling rate, and PAPR. For details about further RF performance tuning, contact RFIC vendors.*

4 Open-loop QEC calibration

The following subsections provide an example of running TRX DC/LO calibration and generating QEC coefficients by running the NXP open-loop QEC calibration tool.

4.1 Enable FR1_FR2_Test_Tool

To enable FR1_FR2_Test_Tool, run the following commands.

```
cd fr1_fr2_test_tool/
./boot_vspa.sh ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD/
```

4.2 Initiate channel

To initiate the channel, run the following command.

```
** ./channels_start.sh <dcsid> **
./channels_start.sh dcs2  *** Calibration on DCS2 only ***

OR

./channels_start.sh  *** Calibration on all channels ***
```

4.3 Check antenna mapping

Check antenna mapping for TX and RX as follows.

```
Ant Mapping TX:
ant_id  dcsid         dcs           status
0       0             LSDCS0-0     RUNNING
1       1             LSDCS0-1     RUNNING
2       2             LSDCS1-0     RUNNING
3       3             LSDCS1-1     RUNNING

Ant Mapping RX:
ant_id  dcsid         dcs           status
0       0             LSDCS0-0     RUNNING
1       1             LSDCS0-1     RUNNING
2       2             LSDCS1-0     RUNNING
3       3             LSDCS1-1     RUNNING
```

4.4 Initiate RF driver DPD LOOP mode

Initiate the RF driver DPD LOOP mode using the below command.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 10
```

4.5 TX IQ QEC

Note: *Be careful when adjusting the QEC tuning values, as improper inputs could potentially lead to boosted power and cause damage to the PA.*

You can fine-tune the DC samples by adjusting the DC offset and IQ imbalance values. However, this step is optional.

```
./update_qec_coeff_tx.sh [tx_ant_id] [imb=gain:phase] [gain=re:im] [dc=re:im]
 [inc]
```

Example:

```
./update_qec_coeff_tx.sh 0 imb=0.23:-0.46 dc=-.00796:.00537
```

You can monitor the constellation on VSA and try to align the constellation of DC samples with non-DC samples.

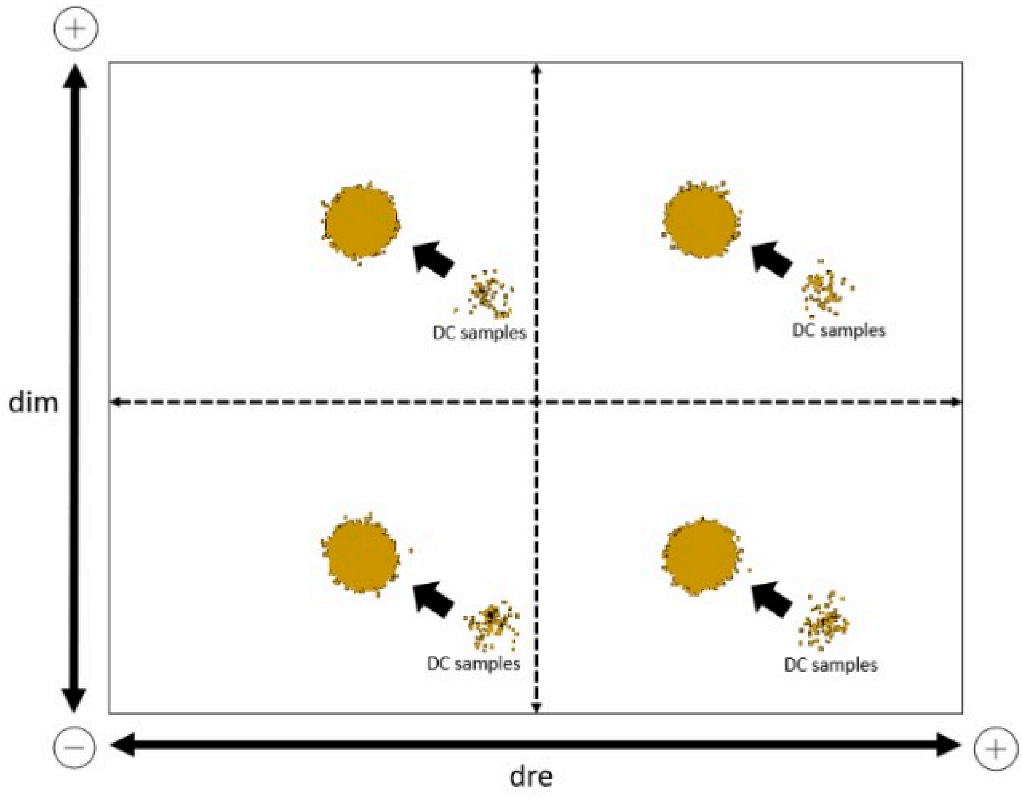

Figure 3. Constellation on VSA

You can directly use the IQ imbalance reported by the VSA as a QEC input data.

The sign symbol should be changed for the QEC command.

For example:

- VSA report:
  - Gain imbalance: 0.23 dB
  - Quad error: -0.46 degree
- QEC command input:
  - --sg -0.23
  - --sp 0.46

Channel Power	20.25 dBm
Channel Power (Active)	20.25 dBm
EVM	5.40 %
EVM Peak	103.82 %
Frequency Error	1190.11 Hz
Symbol Clock Error	0.402 ppm
IQ Offset	-35.57 dB
Time Offset	-51 ns
Sync Correlation	99.9 %
Sync Source	PDSCH DMRS
Magnitude Error	3.68 %
Phase Error	0.04 rad
Gain Imbalance	0.23 dB
Quad Error	-0.46 degree
Timing Skew	-40.19 ps

Figure 4. Demodulation error summary

4.6 RX IQ QEC – IQ imbalance

The IQ imbalance numbers of phase and gain error should be retrieved in single tone mode. The waveform source can be from an external signal generator or DAC via a loopback path.

The steps covered in subsequent subsections are based on LOOPBACK mode.

4.7 Send single tone 12.88 MHz

Send a single tone at 12.88 MHz using the following command.

```
./send_single_tone.sh <tx_ant_id> 12288000 20
```

4.8 Retrieve IQ imbalance numbers

To retrieve IQ imbalance numbers, run the following commands.

```
./update_qec_coeff_rx.sh 0 dis
./qec_open_loop_rx.sh <tx_ant_id> <rx_ant_id>
Calulate IQ imbalance ...
Phase_error = -0.44547
Gain_error = 0.17959
./update_qec_coeff_rx.sh 0 imb=0.17959:-0.44547
RX QEC coeff for antenna 0 on core 1 updated from address 0x23616c0000
Updated coeff saved to file ./qec/qec_coeff_rx_ant0.bin
```

4.9 Disable single tone mode

Disable the single tone mode using the following command.

```
./send_single_tone.sh <tx_ant_id> dis
```

4.10 RX IQ QEC – DC offset

You can further fine-tune the RX DC offset by means of IQ correction.

Section 4.11 and IQ constellation correction provide an example of how to measure the RX performance and run RX IQ correction.

4.11 RX measurement: Convert RX dump .bin to Keysight format

```
./dump_time_domain_rx.sh 0
python3 utils/conv_bin2csv.py -i rx_timedomain_20ms_245760ksps_dump_ant0.bin -d
 td -m fr1

Loading... rx_timedomain_20ms_245760ksps_dump_ant0.bin
>> TD Waveform sampling rate 245760000
resample from 245760000 to 122880000
1228800
>> Original TD waveform length in 20 slots = 1228800
>> Output .csv location =
 rx_timedomain_20ms_245760ksps_dump_ant0_20231103013515.csv
```

On Keysight VSA

Recall >> Recording >> Recall From >> select .csv file

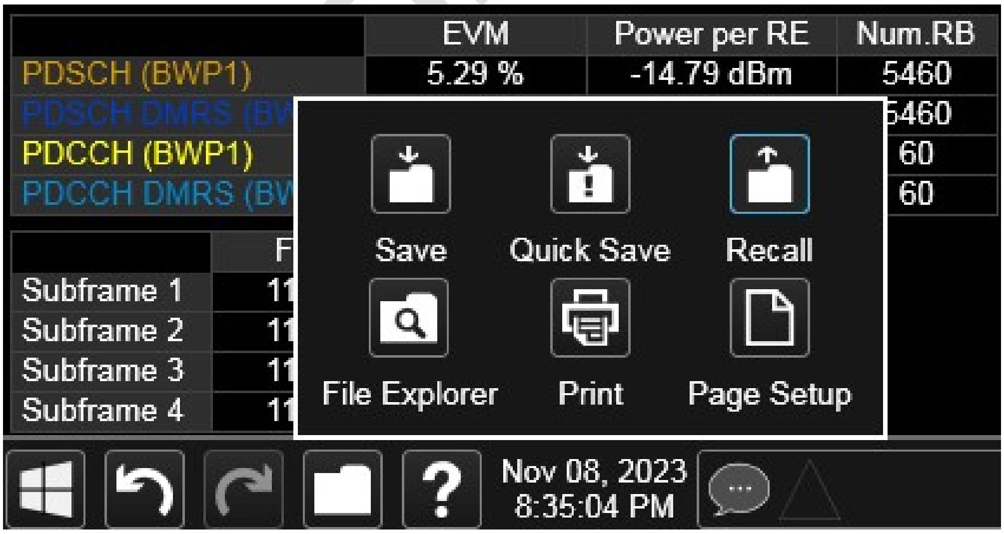

Figure 5. VSA Recall configuration

Meas Standard >> RB Alloc Preset >> DL NR-TM1.1 (P0) >> Apply Preset

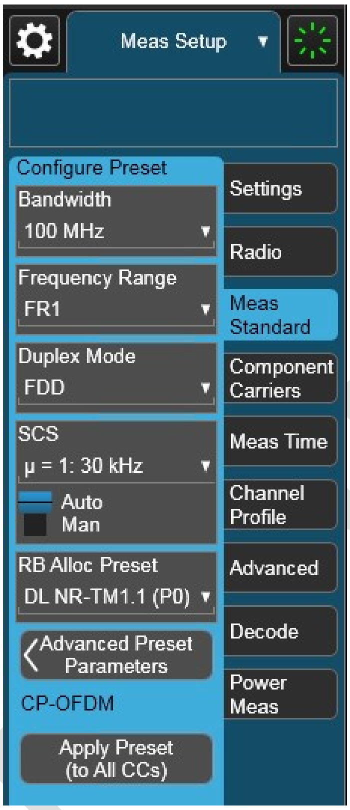

Figure 6. VSA Meas setup

In the following VSA snapshot, the RX EVM rms is 8.76 % and the EVM peak is around 280 %.

OFDM samples affected by DC leakage cause distortion in constellation and degradation in DPD performance.

Figure 7. RX EVM before calibration

4.12 IQ constellation correction

Note: *The current IQ correction only supports the FR1 TM1.1 waveform.*

```
*** ./rx_iq_correction.sh <tx_ant_id> <rx_ant_id>***

./rx_iq_correction.sh 0 0
Apply QEC Rx passthrough mode on DC offset values...
./update_qec_coeff_rx.sh 0 dc=0:0 inc
Starting Rx IQ correction...
incr_scale is set to default 4...
IQ delta = 190336
./update_qec_coeff_rx.sh 0 dc=-.04226:.02186 inc
IQ delta = 2256
./update_qec_coeff_rx.sh 0 dc=-.04170:.02186 inc
IQ delta = 1049
./update_qec_coeff_rx.sh 0 dc=-.04147:.02186 inc
IQ delta = 580
calibration done...found the best --dre -.04147 --dim .02186
```

After RX QEC calibration, QEC coefficients are automatically applied in VSPA.

All generated QEC coefficient binaries for both TX and RX are stored in the following files.

- qec/qec_coeff_rx_ant<rx_ant_id>.bin
- qec/qec_coeff_tx_ant<tx_ant_id>.bin

4.13 Recheck calibrated RX

Repeat the step mentioned in Section 4.11 to play the RX dump on Keysight VSA.

The RX EVM peak is further improved from 280 % to 24 %.

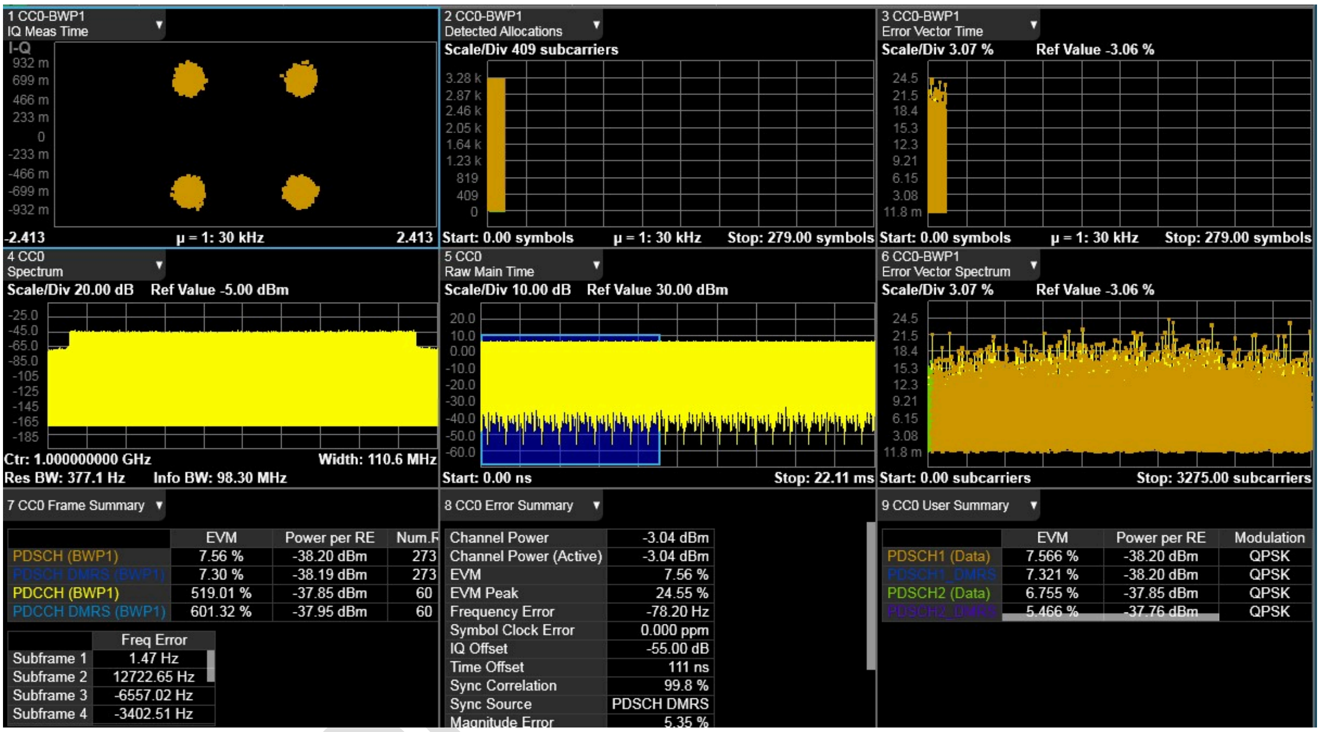

Figure 8. RX EVM after calibration

5 Closed-loop DPD training from time-domain waveform source

5.1 Enable FR1_FR2_Test_Tool

To enable FR1_FR2_Test_Tool, run the following command.

```
./boot_vspa.sh
 ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
```

5.2 Initiate channel

Initiate the channel using the following command.

```
** ./channels_start.sh <dcsid> **

./channels_start.sh dcs2  *** LS Training with observation path only on DCS2:
 LS-ADC ***

OR
```

```
./channels_start.sh dcs2 dcs5  *** HS Training with observation path DCS4: HS-
ADC ***
```

5.3 Check antenna mapping

Check antenna mapping for TX and RX as follows.

```
Ant Mapping TX:
ant_id  dcsid         dcs           status
0       2             LSDCS1-0     RUNNING

Ant Mapping RX:
ant_id  dcsid         dcs           status
0       2             LSDCS1-0     RUNNING
```

5.4 Initiate RF driver DPD LOOP mode

To initiate RF driver DPD LOOP mode, run the following command.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 10
```

5.5 Apply QEC coefficient

Apply QEC TX and RX coefficients generated by QEC calibration.

```
./update_qec_coeff_tx.sh <tx_ant_id> qec/qec_coeff_tx_ant<tx_ant_id>.bin; *** Tx ***
./update_qec_coeff_rx.sh <rx_ant_id> qec/qec_coeff_rx_ant<rx_ant_id>.bin;  *** Rx ***
```

5.6 Adjust output signal power

Run the following command to adjust output signal power (if needed).

```
./scale_percent.sh <tx_ant_id> 140
```

5.7 Customize DPD model

You can choose one of the preset DPD models in config.dat. The following example shows how to use the customized DPD model R&D1 for DPD training. Note that only p can be changed in the r/p/q mapping.

```
use_custom_dpd_model=1  #valid only for T4x image

#This model is used by ADvspa_images_LS.2T2R.T4x_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD
dpd_model_para_custom=(
#r q p(max)             example: 73   522  R&D1 R&D2 R&D2M
 0 0  6  #max 6                  6     4    6    5    5
 1 0  3  #max 5                  -     4    3    2    2
 2 0  -  #max 3                  -     -    -    -    -
 3 0  -  #max 3                  -     -    -    -    -
 4 0  -  #max 3                  -     -    -    -    -
 5 0  -  #max 3                  -     -    -    -    -
 0 1  5  #max 5                  -     4    5    4    4
 1 1  5  #max 6                  6     4    5    3    4
 2 1  -  #max 5                  -     -    -    -    -
 3 1  -  #max 5                  -     -    -    -    -
 4 1  -  #max 5                  -     -    -    -    -
 5 1  -  #max 5                  -     -    -    -    -
 0 2  -  #max 5                  -     -    -    3    4
 2 2  -  #max 6                  6     -    -    3    3
 0 3  -  #max 5                  -     -    -    -    -
 3 3  -  #max 5                  -     -    -    -    -
 0 4  -  #max 5                  -     -    -    -    -
 4 4  -  #max 5                  -     -    -    -    -
 0 5  -  #max 5                  -     -    -    -    -
 5 5  -  #max 5                  -     -    -    -    -
)
```

Figure 9. DPD model configuration

5.8 Run DPD training

LS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 0 1  # 1 iteration
```

```
./dpd_training_close_loop.sh 0 0 3  # 3 iterations in a row
```

HS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 4 1  # 1 iteration
```

```
./dpd_training_close_loop.sh 0 4 3  # 3 iterations in a row
```

Check the training results and record the log.

```
root@fr1_fr2_test_tool# ./dpd_training_close_loop.sh 0 0 1
Ant 0 is set active.
DPD model: [ 0 0 6 ][ 1 0 3 ][ - - - ][ - - - ][ - - - ][ - - - ][ 0 1 5 ][ 1 1 5 ][ - -
 - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ -
 - - ][ - - - ]
Dumping DPD output to file tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
```

```
Using memory from 0xa06d5e8000 size 98304 as intermediate buffer for dumping...
Dumping feedback to file rx_timedomain_35840_245760ksps_dump_ant0.bin
Using memory from 0xa06d5f0000 size 65536 as intermediate buffer for dumping...

*****************************DPD CLOSE LOOP TRAINING RUN COUNTER = 1
 **************************************
Ant 0 is set idle before training.
./dpd_training/dpdt ./dpd_training/dpd_spec.cfg 16384
 tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
 rx_timedomain_35840_245760ksps_dump_ant0.bin 1 0.00001
dpdt 1.7 (Aug 14 2023)
Copyright 2023 NXP
DPD model (GMP polyphase) structure (initialized from file ./dpd_training/dpd_spec.cfg):
  Number of internal phases (R): 1
  GMP description: signal lag (r), envelope lag (q), poly order (p):

  #:  0  1  2  3
 ----------------
  r:  0  1  0  1
  q:  0  0  1  1
  p:  6  3  5  5

SRX measurements:
        avg power = 0.024804 = -16.0548 dBFS
        peak power = 0.230119 = -6.38047 dBFS
        PAPR = 9.6743 dB
Sync:.........
        54.6641 samples offset between ref and srx
        Level alignment gain: 1.50896
Diagonal loading: 1e-05
Training NMSE: -27.4315 dB
Training complete, coefficients written to dpd_coeff_vspa.flp
Ant 0 is set active after training.
TX signal shut down before applying DPD coeff.
Converting DPD coeff struct to MDPD required struct.
Updating DPD coeff from file on antenna 0 from address 0x23616c1000 to vspa
Current DPD model 19: [ 0 0 6 ][ 1 0 3 ][ - - - ][ - - - ][ - - - ][ - - - ][ 0 1 5 ][ 1
 1 5 ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ][ - - - ]
[ - - - ][ - - - ][ - - - ]
Dumping DPD output to file tx_timedomain_69632_DPDoutput_491520ksps_dump_ant0.bin
Using memory from 0xa06d5e8000 size 98304 as intermediate buffer for dumping...
TX signal resumed.
Dumping feedback to file rx_timedomain_35840_245760ksps_dump_ant0.bin
Using memory from 0xa06d5f0000 size 65536 as intermediate buffer for dumping...

DPD Training done.
```

5.9 DPD model searching

FR1_FR2_Test_Tool provides an approach to search a DPD model depending on different board designs or PA types. The results are recorded including DPD model r/q/p mapping, ACLR numbers, and estimated core load, which helps to deal with the trade-off on different system design. The chosen DPD model can be applied manually in config.dat for further adjustment.

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration> srch
```

Example:

```
./dpd_training_close_loop.sh 0 0 1 srch  # 1 iteration
```

The searching process takes some time, ranging from several minutes to even half an hour. The duration of the process depends on the number of polynomials and the desired iterations.

Figure 10. DPD auto search result

```
aclr=-38.07 aclrmpy100=-3807
Previous DPD model: [ 0 0 3 ][ 1 0 4 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 5 ][ 1 2 5 ][ 2 2 6 ]

END of DPD model searching ROUND 1, total searched num of models 49. Result saved in file ./dpd_training/dpd_model_search_report2023
09130722
Best DPD model ROUND 1:  [ 0 0 3 ][ 1 0 4 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 5 ][ - - - ][ 2 2 6 ] ACLR=-38.42 Est_core_load
 50%
Please choose next step:
1 Continue next ROUND of searching,
0 Stop searching and apply the best model and check ACLR on equipment.
You choice:
```

At the end of the searching process, you can choose to:

Press 1 to continue another searching.

Or

Press 0 to stop and train with the best model.

The searching process continues round 2 after 20 s if there is no input for the next step choice.

The printed ACLR number in the report is derived from the ADC dump .bin file. It could be lower than the VSA reported numbers due to the impact of RX gain/attenuation/filter. You can find the report in the dpd_training/ folder.

```
dpd_model_searched_idx_122=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ - - - ][ - - - ][ 2 2 4 ] ACLR_122=-37.62 Est_core_load 41%
dpd_model_searched_idx_123=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 1 ][ - - - ][ 2 2 4 ] ACLR_123=-38.48 Est_core_load 44%
dpd_model_searched_idx_124=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 5 ] ACLR_124=-39.16 Est_core_load 46%
dpd_model_searched_idx_125=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 6 ] ACLR_125=-38.92 Est_core_load 47%
dpd_model_searched_idx_126=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ - - - ] ACLR_126= -37.9 Est_core_load 39%
dpd_model_searched_idx_127=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 1 ] ACLR_127=-38.88 Est_core_load 41%
dpd_model_searched_idx_128=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 2 ] ACLR_128=-38.99 Est_core_load 42%
dpd_model_searched_idx_129=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 3 ] ACLR_129= -38.9 Est_core_load 44%
dpd_model_searched_idx_130=[ 0 0 3 ][ 1 0 5 ][ 2 0 1 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_130=-38.87 Est_core_load 47%
dpd_model_searched_idx_131=[ 0 0 3 ][ 1 0 5 ][ 2 0 2 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_131= -38.8 Est_core_load 49%
dpd_model_searched_idx_132=[ 0 0 3 ][ 1 0 5 ][ 2 0 3 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_132=-38.94 Est_core_load 50%
dpd_model_searched_idx_133=[ 0 0 3 ][ 1 0 5 ][ 2 0 4 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_133=-39.01 Est_core_load 51%
dpd_model_searched_idx_134=[ 0 0 3 ][ 1 0 5 ][ 2 0 5 ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_134=-38.62 Est_core_load 52%
dpd_model_searched_idx_135=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 1 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_135= -39.0 Est_core_load 47%
dpd_model_searched_idx_136=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 2 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_136= -38.9 Est_core_load 49%
dpd_model_searched_idx_137=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 3 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_137=-38.93 Est_core_load 50%
dpd_model_searched_idx_138=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 4 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_138= -38.9 Est_core_load 51%
dpd_model_searched_idx_139=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ 2 1 5 ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR_139=-39.06 Est_core_load 52%
dpd_model_searched_idx_140=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 1 ][ 2 2 4 ] ACLR_140=-38.88 Est_core_load 47%
dpd_model_searched_idx_141=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 2 ][ 2 2 4 ] ACLR_141=-38.96 Est_core_load 49%
dpd_model_searched_idx_142=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 3 ][ 2 2 4 ] ACLR_142=-38.99 Est_core_load 50%
dpd_model_searched_idx_143=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 4 ][ 2 2 4 ] ACLR_143=-38.98 Est_core_load 51%
dpd_model_searched_idx_144=[ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ 1 2 5 ][ 2 2 4 ] ACLR_144=-38.73 Est_core_load 52%
Best DPD model ROUND 1:  [ 0 0 3 ][ 1 0 5 ][ - - - ][ 0 1 2 ][ 1 1 2 ][ - - - ][ 0 2 2 ][ - - - ][ 2 2 4 ] ACLR=-39.25 Est_core_load 45%
root@fr1_fr2_test_tool# ~
```

Figure 11. DPD result log

5.10 PAPR measurement

The PAPR of TM1.1_100MHz_30kHz_FDD_timedomain_491520ksps.bin is 9.68 dB. Since the nonlinearity of PA causes gain compression in higher input power, the measured PAPR of SRx could be compressed.

You can monitor the CCDF curve on VSA and compare the PAPR between TX waveform and SRx waveform to make sure the linearity over the DPD training process.

```
python utils/calc_papr.py -i tx_timedomain_20ms_491520ksps_dump_ant0.bin

Input file is " tx_timedomain_20ms_491520ksps_dump_ant0.bin
papr before CFR: 8.93
papr before CFR: 9.51 db
```

Note:

- *Proper RX gain setting on RF is needed for correct PAPR observation.*
- *ADC clipping could cause the observed PAPR to be lower than the actual value.*

6 Closed-loop DPD training from frequency-domain waveform source

Note:

- *Customize DPD model is not supported in frequency domain mode.*
- *DPD model searching is not supported in frequency domain mode.*

6.1 Enable FR1_FR2_Test_Tool

To enable FR1_FR2_Test_Tool, run the following command.

```
*** LS Training with observation path on LS-ADC ***
./boot_vspa.sh ADvspa_images_LS.4T4R_100M_30K_491_245_TDDFDD/

OR

*** HS Training with observation path on HS-ADC ***
./boot_vspa.sh
 ADvspa_images_LS.2T2R_100M_30K_491_245_HS.2R_400M_120K_983_983_TDDFDD/
```

6.2 Initiate channel

Initiate channel using the following command.

```
** ./channels_start.sh <dcsid> **
./channels_start.sh  *** LS Training with observation path on LS-ADC ***

OR

./channels_start.sh dcs2  *** LS Training with observation path only on DCS2:
 LS-ADC ***

OR

./channels_start.sh dcs2 dcs4  *** HS Training with observation path on DCS4:
 HS-ADC ***
```

6.3 Check Antenna mapping

Check antenna mapping for TX and RX as follows.

```
Ant Mapping TX:
ant_id  dcsid        dcs           status
0       0            LSDCS0-0    RUNNING
1       1            LSDCS0-1    RUNNING
2       2            LSDCS1-0    RUNNING
3       3            LSDCS1-1    RUNNING

Ant Mapping RX:
ant_id  dcsid        dcs           status
```

```
0        0            LSDCS0-0    RUNNING
1        1            LSDCS0-1    RUNNING
2        2            LSDCS1-0    RUNNING
3        3            LSDCS1-1    RUNNING
```

6.4 Initiate RF driver DPD LOOP mode

To initiate RF driver DPD LOOP mode, run the following command.

```
./<path>/diora_ls_dpd_3qb2_trx1.sh ** Refer to Section 10
```

6.5 Apply QEC coefficient

Apply the QEC TX and RX coefficients generated by the QEC calibration.

```
./update_qec_coeff_tx.sh <tx_ant_id> qec/qec_coeff_tx_ant<tx_ant_id>.bin; *** Tx ***
./update_qec_coeff_rx.sh <rx_ant_id> qec/qec_coeff_rx_ant<rx_ant_id>.bin;  *** Rx ***
```

6.6 Run DPD training

LS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 0 1  # 1 iteration
```

```
./dpd_training_close_loop.sh 0 0 3  # 3 iterations in a row
```

HS training

```
./dpd_training_close_loop.sh <TX_ant_id> <RX_ant_id> <iteration>
```

Examples:

```
./dpd_training_close_loop.sh 0 4 1  # 1 iteration
```

```
./dpd_training_close_loop.sh 0 4 3  # 3 iterations in a row
```

7 DPD open-loop training

The open-loop DPD training can be a useful tool for engineering development or in situations where there is no DPD loopback path on the RF side. It is recommended to use a high sampling rate of instrument configuration when conducting the open-loop DPD training.

Requirements:

- A 10 ms pulse trigger is needed to synchronize between LA1224-RDB-B and VSA.
- Connect J526 connector to VSA trigger input.

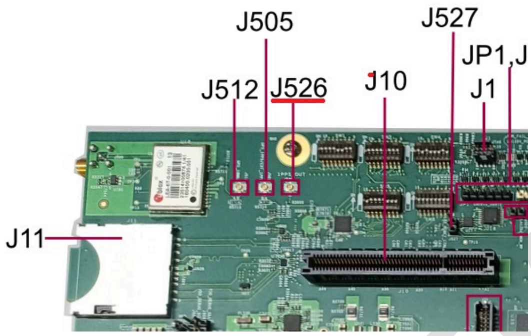

Figure 12. J526 location on LA1224-RDB-B

- Select the trigger source on VSA.

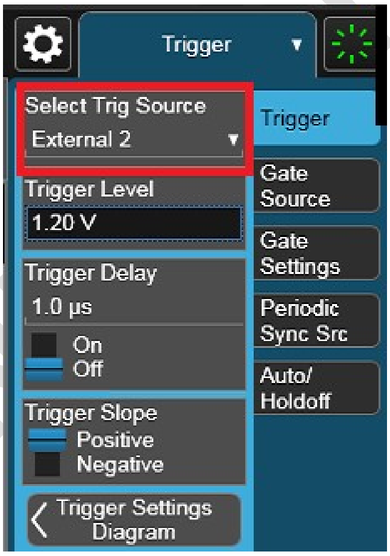

Figure 13. VSA trigger source selection

7.1 Enable FR1_FR2_Test_Tool and RF

Enable FR1_FR2_Test_Tool and RF. For details, run the steps mentioned in Section 5.1 to Section 5.4.

7.2 Capture PA out data from VSA and record as .binx

Save the captured recording as .binx file as follows.

Save >> Recording >> Save As >> All Supported Files >> .binx >> Save >> Recording_0000.binx

```
dd if=Recording_0000.binx of=Recording_0000.bin bs=512 count=1024
```

7.3 Generate DPD coefficient

Generate a DPD coefficient using the following commands.

```
./dpd_training_open_loop.sh <ant_id> Recording_0000.bin <sample rate>
./dpd_training_open_loop.sh 0 Recording_0000.bin 245760
```

8 Acronyms

Table 1 lists the acronyms used in this document.

Table 1. Acronyms

Term	Definition
5G	Fifth-generation mobile communications
ADC	Analog-to-digital converter
BSP	Board support package
DAC	Digital-to-analog converter
DCS	Data conversion subsystem
DPD	Digital predistortion
eCPRI	Enhanced common public radio interface
GPS	Global positioning dystem
HS DCS	High-speed data conversion system
IMRR	Image rejection ratio
LO	FR1 center frequency
LS DCS	Low-speed data conversion system
NIC	Network interface controller
OFDM	Orthogonal frequency-division multiplexing
PA	Power amplifier
PCIe	Peripheral component interconnect express
QEC	Quadrature error correction
RF	Radio frequency
RX	Receiver
TDD	Time division duplex
TX	Transmitter
VSA	Vector signal analysis
VSG	Vector signal generation
VSPA	Vector Signal processing accelerator

9 Related documentation

Table 2 lists and explains the additional documents and resources that you can refer to for more information on LA12xxRDB and LA12xx-SDK. Some of the documents listed below may be available only under a non-disclosure agreement (NDA). To request access to these documents, contact your local NXP field applications engineer (FAE) or sales representative.

Table 2. Related documentation

Document	Description	Link / how to access
LA1224 Reference Design Board Getting Started Guide	Explains the LA1224RDB board settings and physical connections that are required to boot the board.	Contact NXP FAE / sales representative

Table 2. Related documentation...*continued*

Document	Description	Link / how to access
LA1224 Reference Design Board Reference Manual	Provides a detailed description of the LA1224RDB board.	Contact NXP FAE / sales representative
LA1224 Reference Design Board (LA1224-RDB-B) Reference Manual	Provides a detailed description of the LA1224-RDB-B board.	Contact NXP FAE / sales representative
LA1224 Reference Design Board (LA1224-RDB-B) Getting Started Guide	Explains the LA1224-RDB-B board settings and physical connections that are required to boot the board.	Contact NXP FAE / sales representative
LA12xx Software Development Kit (LA12xx-SDK) User Guide	This document explains how to use the Layerscape Access LA12xx BSP for LA12xxRDB. It provides the description of U-Boot, Linux, DPAA2 Ethernet interfaces, and FreeRTOS support on the LA12xxRDB and RF Subsystem.	Contact NXP FAE / sales representative

10 Appendix A: Diora 3QB2 RF DPDLOOP sequence example for LS DPD training

- diora_ls_dpd_3qb2_trx1.sh
- CF: 3.5 GHz
- TRX1 only

```
#!/bin/bash
echo 3 >/proc/sys/kernel/printk

function dio () {
                 gul_refapp -c "$@"  >/dev/null 2>&1
                 echo run: $@
                 sleep 0.5
                }

dio "diora-close"
dio 'diora-open 0 0';
dio 'diora-version';
dio 'diora-setls 1';
dio 'diora-resetn 0';
dio 'diora-txrxsw 1';
dio 'diora-resetn 1';
dio 'diora-setpa 3 0';
dio 'diora-setdpd 3 0';
dio 'diora-setlna 3 0';
dio 'diora-txrxsw 1';
dio 'diora-txrxsw2 1';

echo Diora load
dio 'diora-load';
dio "diora-getsysstatus"

dio 'diora-calresistor 8';
dio 'diora-calregulator';
dio 'diora-setrxbw 11';
dio 'diora-settxbw 3';
dio 'diora-settrxpll 1 7000000';
dio 'diora-setpath 5 2 0 1 11 3';
dio 'diora-setgain 4 1 0 1 19';
```

```
dio 'diora-setgain 1 1 0 1 7';
dio 'diora-setactive 1 1';
dio 'diora-txrxsw 0';
dio 'diora-txrxsw2 0';
dio 'diora-setpa 1 1';
dio 'diora-setlna 1 0';
dio 'diora-setdpd 1 1';
dio "diora-getsysstatus"
```

Note: *For command details, contact the RF vendor.*

11 Appendix B: Diora 3QB2 n78 band LS DPD training result

```
DPD model
[ 0 0 5 ][ 1 0 5 ][ - - - ][ 3 0 1 ][ - - - ][ 5 0 3 ][ 0 1 5 ][ 1 1 0 ][ - -
 - ][ 3 1 1 ][ - - - ][ - - - ]
[ - - - ][ - - - ][ 0 3 4 ][ 3 3 4 ][ - - - ][ - - - ][ - - - ][ 5 5 0 ]
```

			Lower				Upper				
			ACP		Reference		ACP		Reference		
	Offs Freq	Integ BW	dBc	dBm	dBm	Car #	dBc	dBm	dBm	Car #	Filter
A	100.000 MHz	98.280 MHz	-44.28	-20.17	24.11	1	-45.36	-21.25	24.11	1	-3 dB
B	200.000 MHz	98.280 MHz	-50.01	-25.91	24.11	1	-49.52	-25.41	24.11	1	-3 dB

Figure 14. ACP view of n78 after DPD training

Figure 15. EVM of n78 after DPD training

12 Appendix C: Diora 3QC n79 band LS DPD training result

```
DPD model
[ 0 0 5 ][ 1 0 5 ][ 2 0 3 ][ - - - ][ - - - ][ 5 0 1 ][ 0 1 4 ][ 1 1 5 ][ - -
 - ][ - - - ][ - - - ][ - - - ]
[ 0 2 5 ][ 2 2 0 ][ 0 3 1 ][ - - - ][ - - - ][ - - - ][ - - - ][ 5 5 0 ]
```

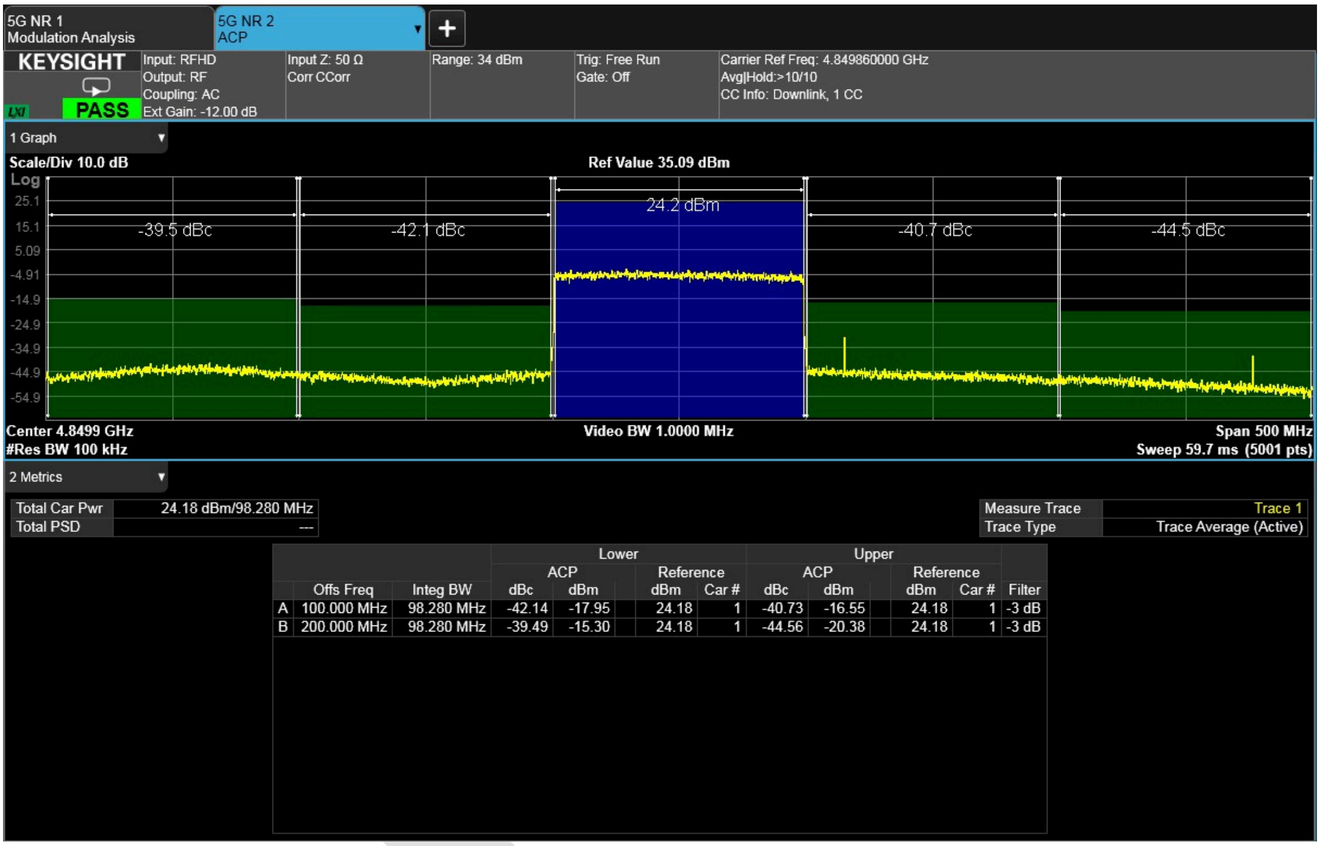

Figure 16. ACP view of n79 after DPD training

Figure 17. EVM of n79 after DPD training

13 Appendix D: RX overall performance after all calibration

- EVM rms: 1.29 %
- EVM peak: 12.62 %
- Gain imbalance: -0.03 dB
- Quad error: -0.19 degree

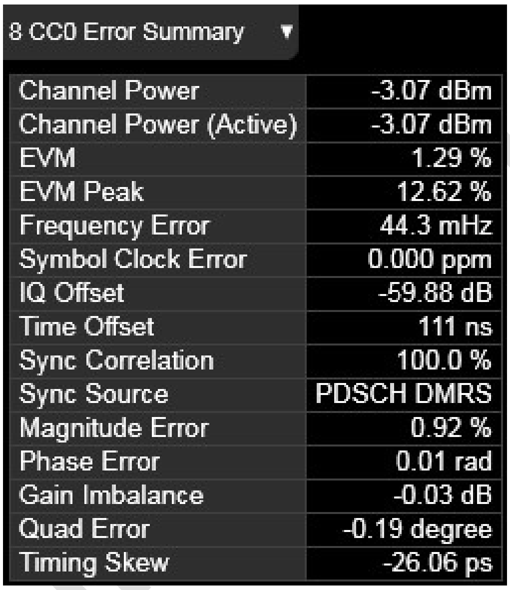

Figure 18. RX error summary

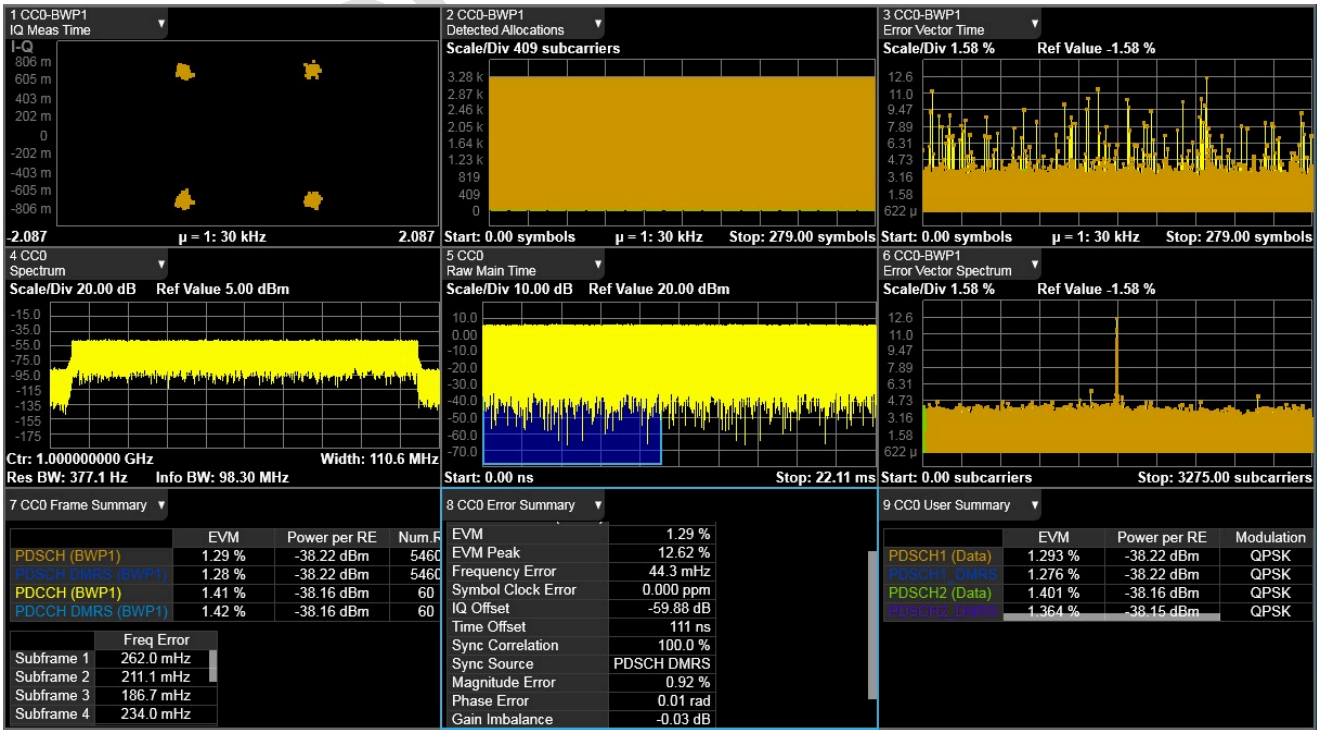

Figure 19. RX EVM

14 Revision history

Table 3 summarizes revisions to this document.

Table 3. Revision history

Document ID	Release date	Description
UG10096 v.A	15 December 2023	Initial NDA release

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